

TENS Machines for Arthritis: Do They Work and How to Use

Complete UK guide to using TENS machines for arthritis pain. How they work, evidence for effectiveness, best devices, settings and NHS availability.

By **Living With Arthritis Editorial Team** · Reviewed by **Clinical Advisory Panel** (HCPC-registered physiotherapy & rheumatology nursing input) · 2026-01-12 · livingwitharthritis.org.uk



*Person using a TENS machine on their leg at home for arthritis pain relief
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TENS (Transcutaneous Electrical Nerve Stimulation) machines are one of the most popular drug-free pain relief options for arthritis in the UK. These small, battery-powered devices deliver mild electrical impulses through electrode pads

placed on the skin near painful joints. With prices starting from just £15 and no prescription needed, TENS machines are accessible and widely used — but do they actually work for arthritis? This guide examines the evidence, explains how to use one effectively and helps you choose the right device.

How TENS Works

TENS machines work through two main mechanisms:

Gate Control Theory

At high frequencies (80–120 Hz), TENS stimulates large nerve fibres that "close the gate" on pain signals travelling to the brain. Essentially, the tingling sensation from the TENS overrides the pain signals, reducing your perception of pain. This is similar to rubbing a bumped elbow — the pressure sensation partially blocks the pain signal.

Endorphin Release

At lower frequencies (2–5 Hz), TENS triggers the release of endorphins — the body's natural painkillers. This takes longer to work (20–30 minutes) but the pain relief can last longer after the device is switched off.

What the Evidence Says



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The evidence for TENS in arthritis is **mixed but generally positive for short-term pain relief**:

- A Cochrane review of TENS for knee osteoarthritis found some evidence of pain reduction, though study quality was variable
- A meta-analysis in Physiotherapy found TENS provided statistically significant pain relief for OA compared to placebo
- NICE acknowledges TENS as an option for osteoarthritis pain management alongside other treatments
- Many rheumatology and physiotherapy departments in the UK recommend TENS as part of multimodal pain management
- Patient satisfaction is generally high — many users report meaningful pain relief even when clinical trial results are modest

TENS is best viewed as one tool in a pain management toolkit, not a standalone treatment. It works well alongside exercise, medication and other approaches.

How to Use a TENS Machine for Arthritis

Pad Placement

Correct electrode placement is crucial for effective pain relief:

- Knee arthritis — Place two pads on either side of the kneecap, or one above and one below the joint
- Hip arthritis — Place pads around the hip joint — one at the front, one at the side or back
- Hand/wrist arthritis — Use smaller pads on either side of the painful joint
- Shoulder arthritis — Place pads on the front and back of the shoulder, or above and below the joint
- Spine/back arthritis — Place pads on either side of the spine at the painful level (never directly on the spine)

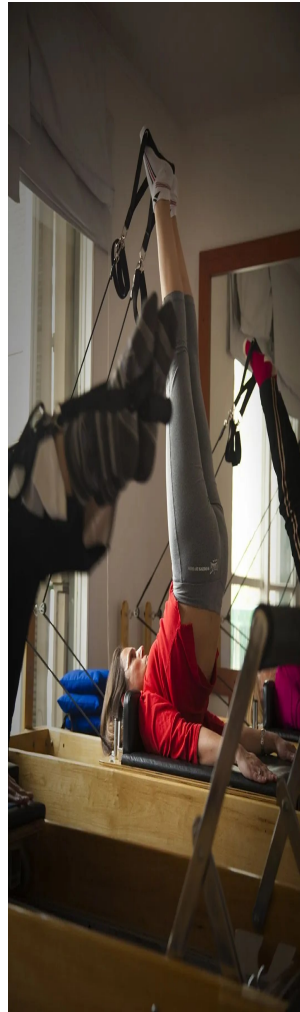
Settings

- Start with a high frequency (80–100 Hz) for immediate tingling pain relief
- Try low frequency (2–5 Hz) for deeper, endorphin-based relief
- Increase intensity gradually until you feel a strong but comfortable tingling — it should never be painful
- Use for 20–60 minutes per session, up to several times daily
- Experiment with settings — what works varies between individuals

Tips for Best Results

- Clean and dry the skin before applying pads
 - Don't place pads on broken skin, over varicose veins or near the front of the neck
 - Replace electrode pads when they lose stickiness (every 20–30 uses typically)
 - Use during activities that usually cause pain (walking, gardening, housework)
 - Try using TENS before exercise to enable more comfortable movement
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Choosing a TENS Machine



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Key features to consider:

- Dual channel — Allows you to use 4 pads simultaneously for larger areas or multiple joints
- Adjustable frequency and intensity — More control over settings for optimal relief
- Rechargeable battery — More economical than disposable batteries
- Pre-set programmes — Useful for beginners, with settings optimised for different pain types
- Timer — Automatic shut-off prevents overuse

Where to Buy in the UK

- Boots, Superdrug and Lloyds Pharmacy stock TENS machines (from £15)
 - Amazon and online retailers offer a wide range
 - Physiotherapy suppliers (e.g., TensCare, Beurer) for more advanced devices
 - Some NHS physiotherapy departments loan TENS machines for trial periods — ask your physiotherapist
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Who Should Not Use TENS

- People with pacemakers or other implanted electrical devices (unless approved by cardiologist)
- During pregnancy (certain pad placements)
- Over areas of reduced sensation (neuropathy)
- On active cancer sites
- While driving or operating machinery

TENS machines are a safe, affordable and drug-free option for arthritis pain relief. While they don't work for everyone, many people find them a valuable addition to their pain management routine. Our virtual assistant can help you with pad placement guidance for your specific joints.

This article is for general information only. Consult your GP or physiotherapist if you're unsure whether TENS is suitable for you.

Q&A

Question: What is the key thing to know about tens machines for arthritis uk: do they work and how to use ?

Short answer: TENS machines are drug-free devices that deliver mild electrical impulses through pads on the skin, a popular choice for arthritis pain relief. They work by stimulating nerves to "close the gate" on pain signals (high frequency, 80-120 Hz) and by triggering endorphin release (low frequency, 2-5 Hz). Evidence suggests short-term pain relief, but they are best used as part of a wider pain management plan.

Question: When should I seek professional advice?

Short answer: If symptoms persist or worsen, pain is severe and unresponsive to usual relief, or you develop new symptoms, contact your GP, physiotherapist or rheumatology team

promptly.

Question: Where can I get free arthritis support in the UK?

Short answer: Arthritis UK: 0800 5200 520. NRAS: 0800 298 7650. Free tools and guides at livingwitharthritis.org.uk.

General information only — not medical advice. Consult your GP or specialist before changing medication or starting a new programme. Emergency: 999. NHS advice: 111. Living With Arthritis · livingwitharthritis.org.uk